

Kaspar 2028 / Development K.ai
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LLM: Model & Backend Benchmarking

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Observation Objective

Meta goal: Realtime speech to speech (thus minimal latency)

- `e2e_ttft` (end to end time to first token) most critical measurement
- `t/s` (token per second) only needs to slightly exceed natural speech (approx. 5 t/s)
 - aim for `15+ t/s` output generation speed

Benchmarking

- hardware: DGX Spark, (RTX 5080 Laptop GPU)
- backends: vllm , sglang , trt-lm , atlas , llama.cpp
- client: llama-benchy
 - 2048 token prompt
 - 128 token response
 - 0 4096 (8192 16384) depth
 - 1 2 concurrency

Power Delivery Issue

additional manual logs via `nvidia-smi`

- `power.draw` (power consumption in watts)
- `clocks.sm` (measured clockspeed in Hz)
- `utilization.gpu` (GPU usage in percent)

Helps to validate that the device is actually running at capacity and not artificially throttling itself due to driver issues. Can be found in the full result files in the [kai-bench repository](#).

TTFT Baseline: Gemma-4-26A4B (d0, pp2048, tg128)

backend	model id	e2e_ttft (ms)	t/s (req)
vLLM	nvidia/Gemma-4-26B-A4B-NVFP4	291.31 ± 2.23	30.53 ± 0.01
SGLang	nvidia/Gemma-4-26B-A4B-NVFP4	487.54 ± 2.14	30.31 ± 0.08
TensorRT-LLM	nvidia/Gemma-4-26B-A4B-NVFP4	499.95 ± 73.43	27.43 ± 0.01
Llama.cpp	unsloth/gemma-4-26B-A4B-it-GGUF	931.75 ± 13.06	49.21 ± 0.03
Atlas	bg-digitalservices/Gemma-4-26B-A4B-it-NVFP4	5402.49 ± 13.28	44.99 ± 0.24

Variant Comparison (vLLM)

variant	e2e_ttft (ms)	t/s (req)
unsloth/gemma-4-E2B-it-NVFP4	139.17 ± 1.16	78.46 ± 0.08
unsloth/gemma-4-E4B-it-NVFP4	213.49 ± 1.23	45.16 ± 0.03
nvidia/Gemma-4-26B-A4B-NVFP4	291.31 ± 2.23	30.53 ± 0.01
unsloth/gemma-4-12B-it-NVFP4	366.62 ± 3.93	21.89 ± 0.01

Quantization Comparison (vLLM)

variant	e2e_ttft (ms)	t/s (req)
unsloth/gemma-4-E2B-it-NVFP4	139.17 ± 1.16	78.46 ± 0.08
google/gemma-4-E2B-it	203.09 ± 1.66	39.89 ± 0.05
unsloth/gemma-4-E4B-it-NVFP4	213.49 ± 1.23	45.16 ± 0.03
google/gemma-4-E4B-it	358.20 ± 0.68	19.60 ± 0.03
nvidia/Gemma-4-26B-A4B-NVFP4	291.31 ± 2.23	30.53 ± 0.01
google/gemma-4-26B-A4B-it	601.42 ± 3.76	23.84 ± 0.01

Impact of Depth on e2e_ttft (vLLM)

variant	d0	d4096	d8192	d16384
unsloth/gemma-4-E2B-it-NVFP4	139.17	442.52	879.76	2144.56
unsloth/gemma-4-E4B-it-NVFP4	213.49	653.67	1285.53	3021.67
nvidia/Gemma-4-26B-A4B-NVFP4	291.31	1004.86	1914.97	4517.58
unsloth/gemma-4-12B-it-NVFP4	366.62	1242.53	2317.21	4956.31

e2e_ttft (ms) per depth

Observations

- **vLLM** outperforms all other tested inference engines on e2e_ttft
- **NVFP4** quantization aids with e2e_ttft (most likely due to memory bandwidth scaling)
- deeper context (eg. very long system prompts) would move us out of realtime capability relatively quickly
- Gemma 4 Unified (12B) variant at **~370** ms blocks significant amount of realtime budget (500ms e2e)
- default llama-benchy warmup is insufficient (strong fluctuations on first run after startup)

Factors impacting e2e_ttft

1. amount of active paramters in model (memory bandwidth)
2. quantization (memory bandwidth)
3. prompt length (tokens preprocessed upon request)
4. depth (preloaded tokens already present in context)
5. concurrency

Next Steps

- Evaluate latency changes for fully on device deployed pipeline (STT -> LLM -> TTS)
- Evaluate latency changes on `image` and `audio` inputs between encoder and unified (12B) variants

Full benchmark logs are provided in [`./resourcen/benchmarks`](./resourcen/benchmarks/). For docker compose files checkout the [`kai-bench`](https://github.com/whatphilipcodes/kai-bench) repository.